

GC University Lahore

BS Computer Science

Semester: 6th

Examination: Final 2023

COURSE CODE: CS4112

TITLE : Introduction to Machine Learning

Time allowed: 155 minutes

Marks: 40

Subjective

Attempt any Four Questions

NOTE Attempt any four 4) questions. Each Question carry equal marks

Question # 1

(a) Define a Support Vector Machine. Also explain the hyper plane and its role in SVM. Marks 05

(b) Discuss different types of hyper plane.

Marks 05

Question # 2

Explain the principle of the back propagation algorithm. Accompany your explanation with a diagram and mathematical model. Explain the use of all the terms and constants that you introduce and comment on the range of values that they can take. Marks 10

Question # 3

Given two coins A and B, with probability of heads being 'p' and 'q' respectively. Initial guess of $p = 0.9$, and $q = 0.5$. Each time one of the coin is chosen at random and tossed, where 5 experiments were done and in each experiment 10 tosses. The following sequence is observed:

Iterations	Sequence of 10 Experiments
1	HTTHTHTHH
2	HHHHTHHHH
3	HTHHHTHTT
4	THHHTHTHT
5	THHHTHHHH

Perform the E Step and M Step on given dataset. Estimate the next values of 'p' and 'q' from the given data. Marks 10

Question # 4

(a) Use the dataset below to learn a decision tree which predicts if people pass machine learning Yes or No, based on their previous GPA High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed
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L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

(b)

What is the entropy $H(\text{Passed})$?

What is the entropy $H(\text{Passed} \mid \text{GPA})$?

What is the entropy $H(\text{Passed} \mid \text{Studied})$? Marks 5

(b) Consider the following dataset for naïve bayes classifier and predict the probability if Bob will default his loan.

Home owner	Marital Status	Job Experience 1 5	Defaulted
Yes	Single	3	No
No	Married	4	No
No	Single	4	No
Yes	Married	2	Yes
No	Divorced	2	Yes
Home owner	Marital Status	Job Experience 1 5	Defaulted
No	Married	2	Yes
Yes	Divorced	2	No
No	Married	3	Yes
Yes	Single	2	Yes

Bob:

Home owner = No

Marital Status = Married

Job Experience = 3

Marks 5

Question # 5

Find three clusters with the help of K-mean algorithm in the following given dataset. Where $k=3$.

X1	Y1
2.2	2.3
3	3.5

4.5	4.8
10.1	12.1
11.2	11.4
12.2	12.8
20.1	21.3
25.2	23.4
30.1	30.4

Marks 10

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